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CSE3016 – AWS SOLUTION ARCHITECT

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SEMESTER: INTERIM SEMESTER 2025-26

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EXP.NO: 01	Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware
DATE: 10-11-25	

AIM

To Launch an EC2 instance using the AWS Management Console and to successfully deploy a website using Apache middle ware

PROCEDURE

1. Launch EC2 Instance: Use the AWS Console to launch a t2.micro instance with Amazon Linux 2023 and an appropriate Key Pair.

2. Configure Security Group: Ensure the instance's Security Group allows inbound traffic on Port 22 (SSH) and Port 80 (HTTP).

3. Connect via SSH: Use your downloaded .pem key file to SSH into the instance using the public IP address.

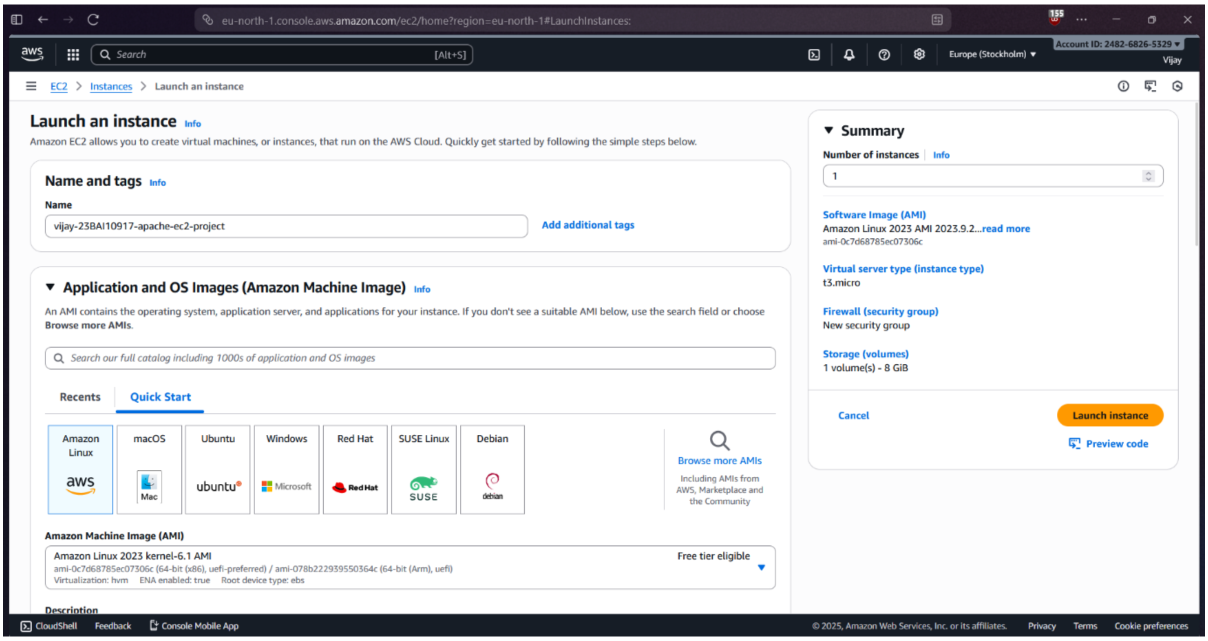
4. Install Apache: Run sudo dnf update -y then sudo dnf install httpd -y on the instance.

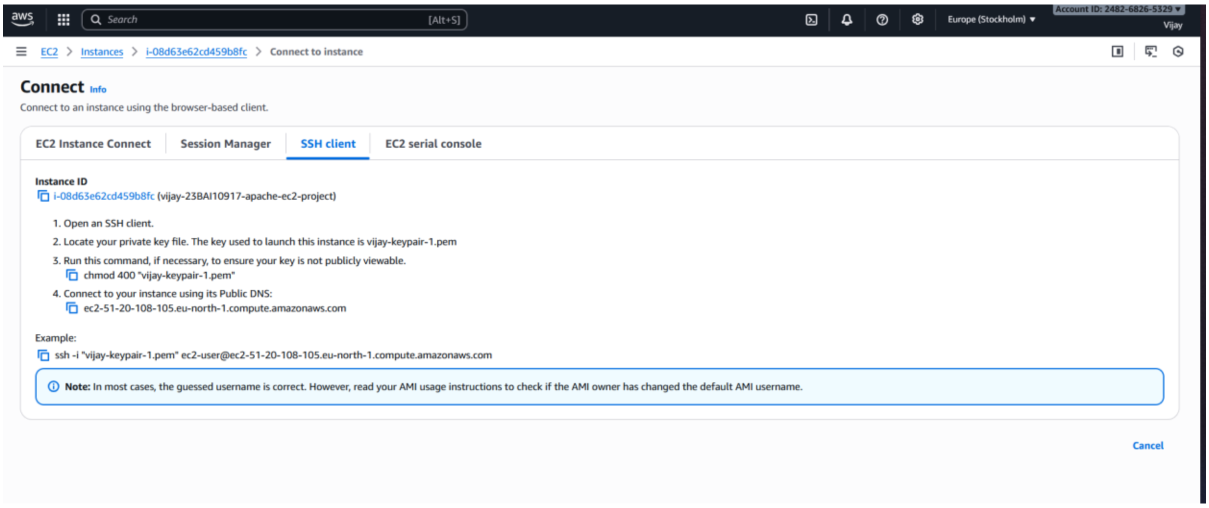
5. Start and Enable Service: Run sudo systemctl start httpd and sudo systemctl enable httpd.

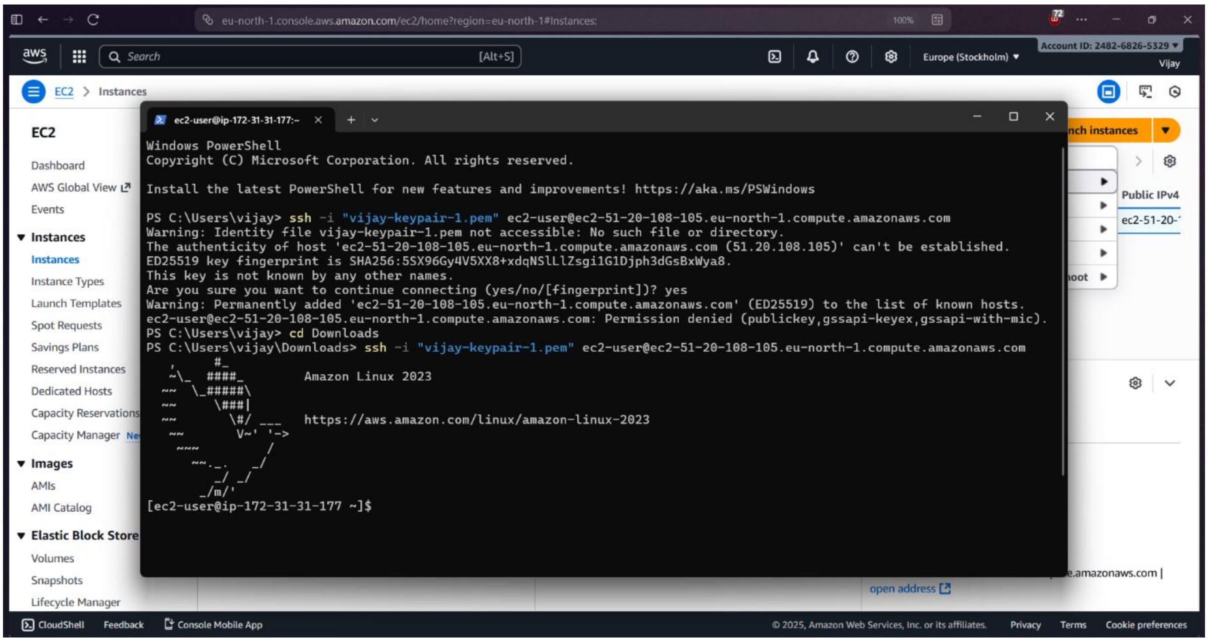
6. Deploy Website: Navigate to /var/www/html/ and create or upload your website files (e.g., sudo nano index.html).

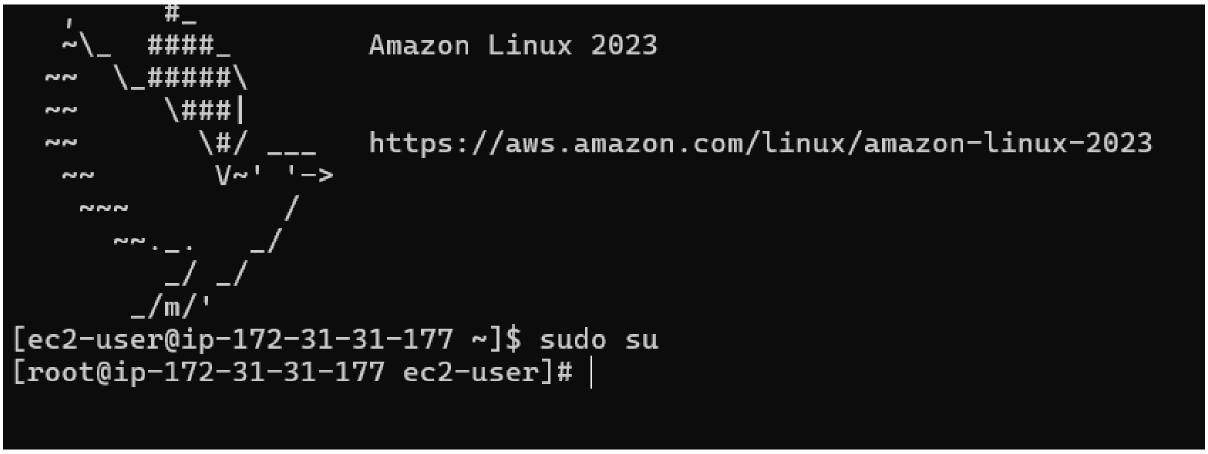
7. Test: Open the instance's Public IPv4 Address in a web browser to view the deployed website.

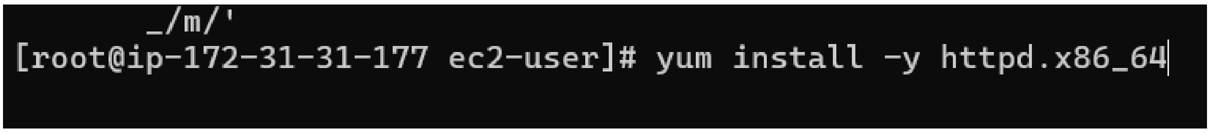
IMPLEMENTATION SCREENSHOT






ec2-user@ip-172-31-31-177:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\vijay> ssh -i "vijay-keypair-1.pem" ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com
Warning: Identity file vijay-keypair-1.pem not accessible: No such file or directory.
The authenticity of host 'ec2-51-20-108-105.eu-north-1.compute.amazonaws.com (51.20.108.105)' can't be established.
ED25519 key fingerprint is SHA256:5SX96Gy4V5XX8+xdqNSlLlZsgi1G1Djph3dGsBxWya8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-51-20-108-105.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
PS C:\Users\vijay> cd Downloads
PS C:\Users\vijay\Downloads> ssh -i "vijay-keypair-1.pem" ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-31-177 ~]$
EC2
Instances
Dashboard
AWS Global View
Events
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Europe (Stockholm)
Public IPv4
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

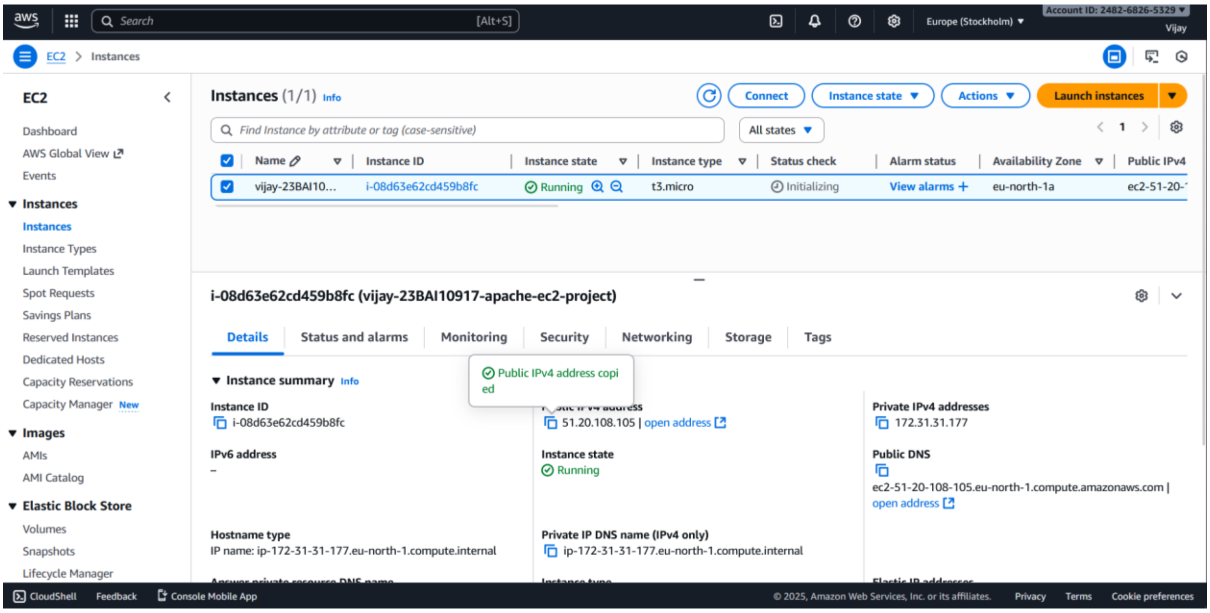

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-31-177 ~]$ sudo su
[root@ip-172-31-31-177 ec2-user]#


[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64

```
[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64
Amazon Linux 2023 Kernel Livepatch repo 273 kB/s |  29 kB     00:00
Dependencies resolved.
==========================================================================
 Package                Arch   Version                    Repository   Size
==========================================================================
Installing:
 httpd                  x86_64 2.4.65-1.amzn2023.0.2      amazonlinux  47 k
Installing dependencies:
 apr                    x86_64 1.7.5-1.amzn2023.0.4       amazonlinux 129 k
 apr-util               x86_64 1.6.3-1.amzn2023.0.2       amazonlinux  97 k
 apr-util-lmdb          x86_64 1.6.3-1.amzn2023.0.2       amazonlinux  13 k
 generic-logos-httpd    noarch 18.0.0-12.amzn2023.0.3     amazonlinux  19 k
 httpd-core             x86_64 2.4.65-1.amzn2023.0.2      amazonlinux 1.4 M
 httpd-filesystem       noarch 2.4.65-1.amzn2023.0.2      amazonlinux  13 k
 httpd-tools            x86_64 2.4.65-1.amzn2023.0.2      amazonlinux  81 k
 libbrotli              x86_64 1.0.9-4.amzn2023.0.2       amazonlinux 315 k
 mailcap                noarch 2.1.49-3.amzn2023.0.3      amazonlinux  33 k
Installing weak dependencies:
 apr-util-openssl       x86_64 1.6.3-1.amzn2023.0.2       amazonlinux  15 k
 mod_http2              x86_64 2.0.27-1.amzn2023.0.3      amazonlinux 166 k
 mod_lua                x86_64 2.4.65-1.amzn2023.0.2      amazonlinux  60 k

Transaction Summary
==========================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-lmdb-1.6.3-1.amzn2023. 369 kB/s |  13 kB     00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64 3.0 MB/s | 129 kB     00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x 2.1 MB/s |  97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn20 737 kB/s |  15 kB     00:00
(5/13): httpd-2.4.65-1.amzn2023.0.2.x86 2.5 MB/s |  47 kB     00:00
(6/13): generic-logos-httpd-18.0.0-12.a 786 kB/s |  19 kB     00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2 578 kB/s |  13 kB     00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0 2.9 MB/s |  81 kB     00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0.  28 MB/s | 1.4 MB     00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2 9.9 MB/s | 315 kB     00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3. 1.3 MB/s |  33 kB     00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0. 6.4 MB/s | 166 kB     00:00
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s |  60 kB     00:00
--------------------------------------------------------------------------
Total                                    14 MB/s | 2.4 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                  1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                  1/13
  Installing       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64        2/13
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_       3/13
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64             4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch             5/13
```

```
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s |  60 kB     00:00
--------------------------------------------------------------------------------
Total                                            14 MB/s | 2.4 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                       1/13
  Installing       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64             2/13
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_            3/13
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64                  4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                  5/13
  Installing       : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64              6/13
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                 7/13
  Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Installing       : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Installing       : httpd-core-2.4.65-1.amzn2023.0.2.x86_64               9/13
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64               10/13
  Installing       : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                 11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3           12/13
  Installing       : httpd-2.4.65-1.amzn2023.0.2.x86_64                   13/13
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64                   13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                       1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                  2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64             3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_            4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3            5/13
  Verifying        : httpd-2.4.65-1.amzn2023.0.2.x86_64                    6/13
  Verifying        : httpd-core-2.4.65-1.amzn2023.0.2.x86_64               7/13
  Verifying        : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Verifying        : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64              9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                 11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64               12/13
  Verifying        : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                 13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-2.4.65-1.amzn2023.0.2.x86_64
  httpd-core-2.4.65-1.amzn2023.0.2.x86_64
  httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
  httpd-tools-2.4.65-1.amzn2023.0.2.x86_64
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-31-177 ec2-user]#
```

```
Complete!
[root@ip-172-31-31-177 ec2-user]# systemctl start httpd.service
[root@ip-172-31-31-177 ec2-user]# systemctl enable start httpd.service
Failed to enable unit: Unit file start.service does not exist.
[root@ip-172-31-31-177 ec2-user]# systemctl enable httpd.service
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-31-177 ec2-user]#
```



Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the Apache HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting www.example.com, you should send e-mail to "webmaster@example.com".

If you are the website administrator:

You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

You are free to use the image below on web sites powered by the Apache HTTP Server:

RESULT

Thus, EC2 instance has been launched using the AWS Management Console and a website is deployed using the Apache Middle ware